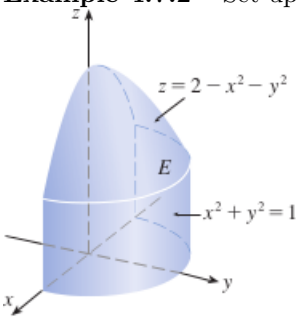
4 Multiple Integrals

4.7 Triple Integrals in Cylindrical Coordinates

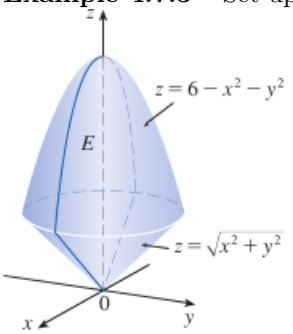
Example 4.7.1 Identify the surface whose equation is

$$r = 2\sin\theta.$$

Example 4.7.2 Set up a triple integral to find the volume of the given shape:



Example 4.7.3 Set up a triple integral to find the volume of the given shape:



Example 4.7.4 Evaluate the integral $\iiint_E x^2 + y^2 \, dV$ where
$E = \{(x, y, z) : -2 \leq x \leq 2, -\sqrt{4 - x^2} \leq y \leq \sqrt{4 - x^2}, \sqrt{x^2 + y^2} \leq z \leq 2\}$.

Example 4.7.5 Find the volume of the solid which lies above the xy-plane and below the surface $z = 4 - x^2 - y^2$.

Example 4.7.6 Evaluate the integral $\iiint_E x\,dV$ where E is enclosed by by the paraboloid $x = y^2 + z^2$ and the plane $x = 4$.

Example 4.7.7 Find the volume of the region bounded between the surfaces $z = \sqrt{x^2 + y^2}$ and $z = 6 - x^2 - y^2$.